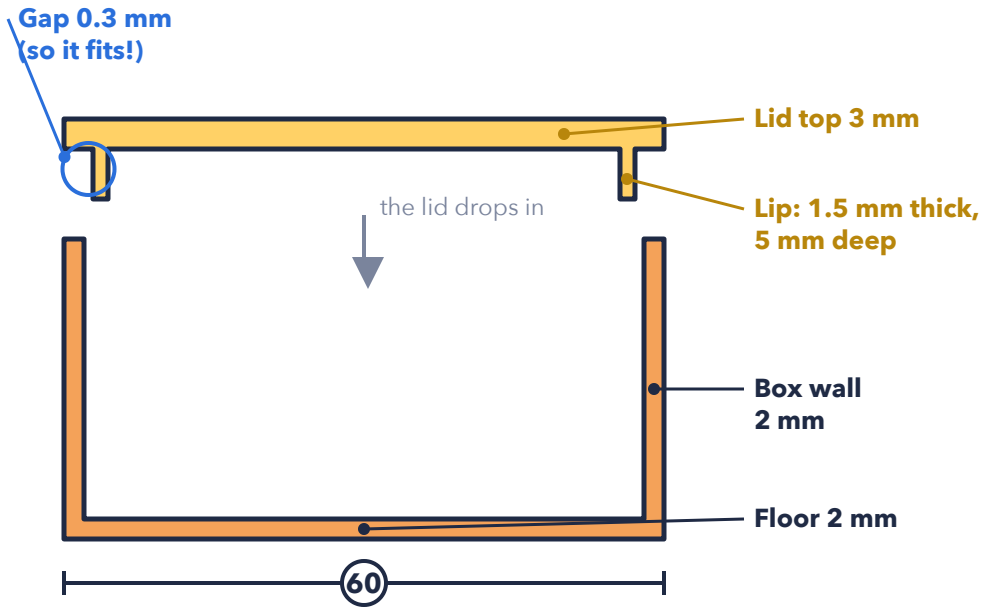
Challenge | OnShape | 2 class periods

Box with a Lid in OnShape

Design a little box with a lid that actually fits. It's a real engineering challenge, because two parts have to fit together!



The box and lid cut in half, so you can see inside. (The gap is drawn bigger than it really is.)

The secret: the lip and the gap

The lid has a **lip**: a thin rim underneath that slides **inside** the box, so the lid can't slide off. But if the lip is exactly the size of the inside of the box, it won't fit! 3D prints come out a tiny bit fat. So we leave a **gap** of `0.3` mm on every side.

BEFORE YOU START

Finish the **Spinning Top** first. Make a new document named **Box – First Name Last Initial**, and set **Workspace units** to **Millimeter**.

1 Sketch the box shape

- [ ] Click **Sketch**, then click the **Top** plane.
- [ ] Click the **Center point rectangle** tool. Click on the **origin** first (that's the center), then click again to make a rectangle.
- [ ] Click **Dimension** (D). Click the top line and type `60`. Click a side line and type `40`.
- [ ] Click the green ✔ to finish the sketch.

2 Pull it up into 3D

- [ ] Click **Extrude**. Click inside your rectangle.
- [ ] Make sure **New** is chosen at the top. Set **Depth** to `30` mm. Click ✔.

> **CHECK**
> You have a solid block, 60 × 40 × 30 mm. It's a brick for now!

3 Round the corners

- [ ] Click **Fillet**. Click the **4 up-and-down edges** at the corners of the block (spin the view to reach the back ones).
- [ ] Set the radius to `5` mm and click ✔.

4 Hollow it out

- [ ] Click **Shell**. For **Faces to remove**, click the **top face** of the block.
- [ ] Set **Thickness** to `2` mm. Click ✔.

> **CHECK**
> Now it's an open box with 2 mm walls and rounded corners. 🎉 The feature list shows: Sketch 1, Extrude 1, Fillet 1, Shell 1.

5 The lid top

We'll copy the outline of the box's rim, so the lid matches perfectly.

- [ ] Click **Sketch**, then click the **top rim** of the box (the thin flat edge all the way around).
- [ ] Click the **Use** tool (U), then click the rim again. OnShape copies its outside **and** inside outlines into your sketch. Press Esc, then click ✔.
- [ ] Click **Extrude**. Click **both** shaded areas of the new sketch: the thin rim ring **and** the big middle.
- [ ] Choose **New** at the top (not Add!). Set **Depth** to **3** mm. If it goes down into the box, click the **flip arrow** so it goes up. Click ✔.

CHECK

The **Parts** list now says **Parts (2)**: the box and the lid. The lid is a flat plate sitting on top.

WHY "NEW"?

"New" makes a separate part. If you pick **Add**, OnShape glues the lid onto the box forever, and you'd have a box you can never open! 😅

6 The lip (the tricky part!)

- [ ] Click **Sketch**, then click the **top rim** of the box again. (The lid is in the way? Click the 👁 eye next to the lid in the Parts list to hide it. Click it again later to show it.)
- [ ] Click **Use** and click the **inside edges** of the box (where the inside walls meet the rim).
- [ ] Click the **Offset** tool. Click the outline you just copied, and type `0.3`. Make sure the new line is on the **inside** (click the flip arrow if it's outside). That's your **gap**.
- [ ] Use **Offset** again on the **same** copied outline, and type `1.8` (that's the 0.3 gap + the 1.5 mm lip). Inside again! Click ✔.
- [ ] Click **Extrude**. Click only the **thin ring between your two offset lines**. That's the lip.
- [ ] Choose **Add** at the top. Set **Depth** to `5` mm, going **down** into the box (flip it if needed).
- [ ] Turn on **Merge scope** and click the **lid**, so the lip joins the lid (not the box!). Click ✔.

CHECK

Show the lid and hide the **box** (eye icon in the Parts list). Flip the view to look under the lid: you should see the lip, a rounded rectangle ring sticking down 5 mm. Then show the box again.

MATH CHECK

Box inside = 60 − 2 − 2 = **56** mm long. Lip outside = 56 − 0.3 − 0.3 = **55.4** mm. That 0.6 mm of total wiggle room is what lets it fit!

7 Make it yours 🎨

- **Change the size:** right-click **Sketch 1** → **Edit**, change 60 or 40, click ✔. Everything else updates by itself, even the lid! (Keep it under `80` mm.)
- **Put your initials on the lid:** **Sketch** on the top of the lid, use the **Text** tool, then **Extrude** it **Add** `1` mm up (merge with the lid).
- **Make it deeper:** edit **Extrude 1** and change 30.

8 Hand it in 📤

- [ ] Check the document name: **Box – First Name Last Initial**.
- [ ] Click **Share**, type your teacher's email (`iggyzs@yahoo.com`), choose **Can view**, turn on **Export**, and click **Share**.
- [ ] Tell your teacher it's ready. The box and lid print as **two separate parts**, and the lid gets flipped over to print.

TEACHER NOTES

Export each part separately (right-click the part in the Parts list → Export → STL). Print the lid upside down (flat top on the bed) so the lip needs no supports. If lids come out too tight on your printer, have students change both offsets: 0.4 / 1.9 instead of 0.3 / 1.8.

🔧 The fit test

When your box is printed, put the lid on. How does it feel? Circle one:

Too tight 😬 Just right 😃 Too loose 🙃

Too tight? Make the gap bigger (both offsets go up by 0.1). Too loose? Make it smaller. Then print the lid again. That's called **iterating**, and real engineers do it all the time!

🤔 Think about it

1. Why did the lid need a gap? What would happen without it?

2. When you changed the box size in Sketch 1, the lid changed too. Why is that useful?